Decision intelligence, not a refund chatbot

Automate the obvious. Investigate the uncertain. Keep humans in control.

A narrow, auditable applied-AI approach to refund operations.

The bottleneck is operational, not technical

BEFORE Legacy approval rule → mixed manual queue → slow low-risk decisions

AFTER Policy engine → safe eligibility OR evidence-rich human review

MANAGER No-code policy draft → impact replay → explicit activation

GOAL / SHORTER QUEUES WITHOUT INCREASING LOSS

Draw the boundary where evidence is strong

AUTO-ELIGIBLE Valid low-risk request; ≤$50; tenure ≥6 months; ≤1 prior refund; policy and evidence satisfied.

HUMAN Fraud, missing evidence, carrier disputes, wrong item verification, discretionary partials, exceptions.

$50 and cohort filters are design assumptions, not quoted source policy.

NO AUTOMATIC REJECTION / NO PAYMENT EXECUTION

A product with two decision workflows

REVIEWER Prioritized queue → case + investigation trace → decision with rationale → audit.

MANAGER Draft a rule → replay history + pending → inspect affected cases and dollars → publish or revert.

BONUS Policy Time Machine makes change impact visible before rollout.

Agentic where useful. Deterministic where critical.

IMPLEMENTED Bounded five-tool read-only investigation workflow + deterministic policy engine.

PROPOSED LLM evidence extractor and specialist Investigation / Policy / Risk agents.

TRUST BOUNDARY Model outputs never independently authorize a payment or override fraud controls.

HONEST SCOPE / NO LIVE LLM IN CURRENT DEMO

Early evidence supports a cautious rollout

PENDING 3 / 21 auto-eligible (14.3%); 18 / 21 human review.

HISTORY 84 / 251 auto-eligible (33.5%); all 84 were previously approved.

IMPORTANT Historical agreement is not accuracy; past decisions and outcomes are noisy.

NEXT GATE / LABELED HOLDOUT + SHADOW MODE

Define success and a kill switch

EFFICIENCY Eligibility rate, minutes saved, time-to-decision, p90 queue age.

SAFETY Wrongful approvals, dollars lost, chargebacks / 1,000, appeals.

ROLLBACK Any critical fraud/evidence violation; >0.5% confirmed erroneous auto approvals; material chargeback rise (proposed triggers).

ILLUSTRATIVE ROI / ~3.35 HOURS SAVED PER WEEK AT 300 REQUESTS

Prove the boundary, then scale the intelligence

NOW Runnable local engine, reviewer, manager simulator, policy replay, tests.

NEXT Annotated evaluation set, shadow mode, durable audit and policy approvals.

THEN Evaluated LLM extraction, limited canary, monitored scale-up.

NOT BUILT Production payments, autonomous LLM agents, hosted deployment.

ASK / APPROVE A SHADOW-MODE PILOT WITH CLEAR STOP CRITERIA